QQ-A-250/10E
September 21, 1978
SUPERSEDING
QQ-A-250/10D
December 9, 1970
(See 6.3)

FEDERAL SPECIFICATION

ALUMINUM ALLOY 5454, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 5454 plate and sheet; the general requirements are covered in QQ-A-250.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as 0, H32, H34 or H112 temper as specified.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements specified in table I.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Magnesium	2.4	3.0
Manganese	0.50	1.0
Chromium	0.05	0.20
Iron	-	0.40
Silicon	-	0.25
Zinc	-	0.25
Titanium	-	0.20
Copper	-	0.10
Others, each	-	0.05
Others, total	-	0.15
Aluminum	Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

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TABLE II. Mechanical properties

Temper	Thickness	Tensile strength		Yield strength 1/ minimum	Elongation in 2 in. or 4 times D 2/, 3/, minimum
		Minimum	Maximum		
	Inches	psi	psi	psi	Percent
O	0.020 thru 0.031	31,000	41,000	12,000	12
	0.032 thru 0.050	31,000	41,000	12,000	14
	0.051 thru 0.113	31,000	41,000	12,000	16
	0.114 thru 3.000	31,000	41,000	12,000	18
H32	0.020 thru 0.050	36,000	44,000	26,000	5
	0.051 thru 0.249	36,000	44,000	26,000	8
	0.250 thru 2.000	36,000	44,000	26,000	12
H34	0.020 thru 0.050	39,000	47,000	29,000	4
	0.051 thru 0.161	39,000	47,000	29,000	6
	0.162 thru 0.249	39,000	47,000	29,000	7
	0.250 thru 1.000	39,000	47,000	29,000	10
H112	0.250 thru 0.499	32,000	-	18,000	8
	0.500 thru 2.000	31,000	-	12,000	11
	2.001 thru 3.000	31,000	-	12,000	15

1/ 0.2 percent offset.
2/ Not required for material 1/2 inch or less in width.
3/ D represents specimen diameter.

4. QUALITY ASSURANCE PROVISIONS (See QQ-A-250).

5. PREPARATION FOR DELIVERY (See QQ-A-250).

6. NOTES

6.1 Intended use. This alloy is intended for use where a weldable moderate strength alloy having comparatively good corrosion resistance is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

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(a) Title, number, and date of this specification.
(b) Form and temper required (see 1.2.1).
(c) Dimensions required.
(d) Requirements for sizes not specifically covered (see QQ-A-250.)
(e) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 Certain provisions of this specification are the subject of international standardization agreement ABC-NAVY-STD-44. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization officer, if required.

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